

UNIVERSIDAD DISTRITAL

Francisco José de Caldas

Computer Engineering Program

Academic Collaboration Network

Skill-Based Study Group Formation & Institutional Resource
Sharing

PROJECT AUTHORS

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Problem Context & Challenges

CURRENT STATE ANALYSIS



Isolated Learning

- ❗ Persistently structural challenge in higher education systems.
- ❗ Inefficient study group formation leads to uneven peer interaction.
- ❗ Students struggle to find peers with complementary skills.



Unbalanced Teams

- ⚠ Spontaneous self-organization creates "Skill Clustering".
- ⚠ High-performers group together, increasing social isolation for others.
- ⚠ Results in homogeneous groups with limited learning potential.



Informal Tools

- ✗ Heavy reliance on non-institutional tools (e.g., WhatsApp).
- ✗ Lack of oversight prevents equitable knowledge distribution.
- ✗ Creates data silos where collaboration metrics are lost.

METHODOLOGY

n = 25

University Students

Primary data collected through a structured survey to validate study group formation barriers.

- ✓ Validated Pain Points
- ✓ Assessed Tech Needs

Platform Dependency



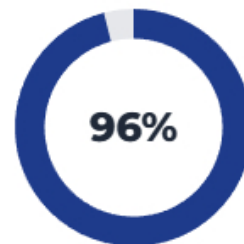
Rely exclusively on **WhatsApp** for academic coordination.

Primary Barrier



Cite **communication or interest deficits** as the main obstacle.

Solution Demand



Expressed strong interest in an **institutional skill-based platform**.

Adoption Likelihood

3.88 / 5.0



High probability of students switching to the proposed system.



Project Objectives

- ✓ Formalize study group formation as a **constrained optimization problem** to remove bias.
- ✓ Mitigate social isolation through **skill-based matching** and institutional integration.
- ✓ Enhance equitable knowledge distribution using **graph-theoretical modeling**.



Methodology

- 📋 **Mixed-Method Approach:** Structured survey (n=25) and observation of current tool usage patterns.
- ⚙️ **Experimental Evaluation:** Simulation using 200 synthetic profiles to test grouping algorithms.
- 📈 **Key Finding:** Complementary variance minimization significantly reduces intra-group

Proposed Platform Concept

SYSTEM ARCHITECTURE & CORE LAYERS

 System Design



Student Profiling Layer

DATA ACQUISITION

- ✓ Technical & Soft Skills Inventory
- ✓ Availability & Schedule Mapping
- ✓ Academic Performance History
- ✓ Learning Style Preferences



Skill Matching Engine

CORE INTELLIGENCE

- ✓ Multi-dimensional Compatibility
- ✓ Complementary Skill Algorithms
- ✓ Variance Minimization ($\text{Min } \Sigma \text{ Var}$)
- ✓ Automated Group Formation



Collaboration Graph

NETWORK MODEL

- ✓ Graph Model: $G = (V, E)$
- ✓ Vertices (V): Student Profiles
- ✓ Edges (E): Collaborative Ties
- ✓ Network Density Analysis



Collaboration Support

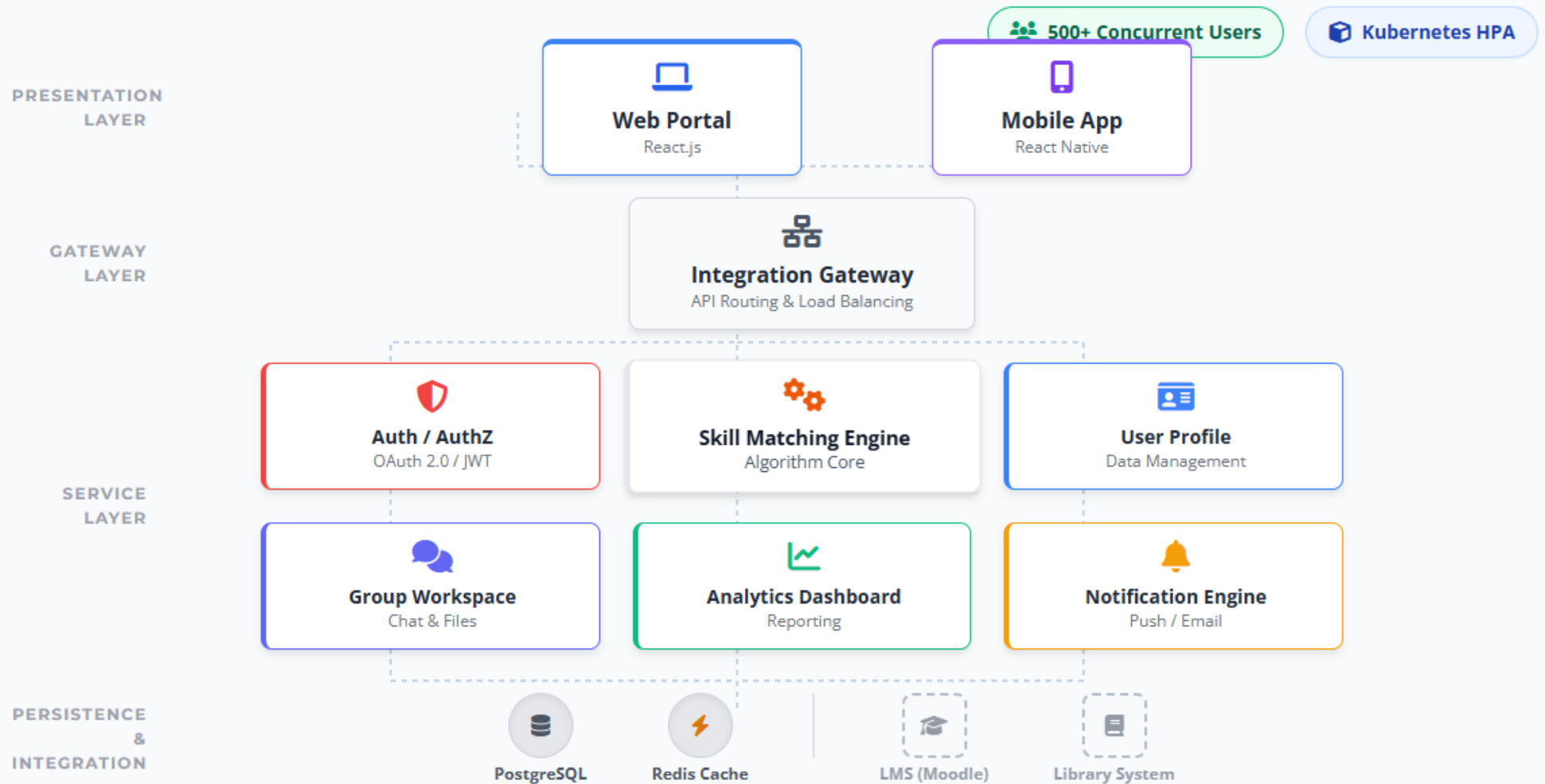
TOOLS & INTEGRATION

- ✓ Digital Group Workspaces
- ✓ Library Resource Linking
- ✓ Institutional LMS Integration
- ✓ Real-time Communication Tools

System Architecture

MICROSERVICES OVERVIEW

Technical Design



Requirements & Tech Stack

SYSTEM SPECIFICATIONS & ARCHITECTURE

 System Specs

 Functional Req.

FR1: Profile Management

High

Complete student profiling with privacy controls.

- ✓ Completeness scoring logic
- ✓ GDPR-compliant data handling

FR2: Intelligent Matching

Critical

Algorithm processing within <3 seconds.

- ✓ Explainable match reasons
- ✓ Multi-dimensional scoring

FR3: Group Workspace

Med

- ✓ Real-time messaging & file sharing (50MB)

FR4-5: Ops & Integration

- ✓ Multi-channel notifications (Push/Email)
- ✓ SSO with LMS & Course Sync

 Non-Functional Req.

 Performance

Page Load
< 2s

API Resp.
< 500ms

 Scale & Reliability

- ✓ Support for 5,000+ total users
- ✓ 99.5% Uptime Guarantee
- ✓ Horizontal Auto-scaling

 Security

TLS 1.3 OAuth 2.0 JWT MFA

 Tech Stack

FRONTEND EXPERIENCE

 **React**
Web Portal

 **React Native**
Mobile Apps

BACKEND SERVICES

 **Node.js**
Core Services

 **FastAPI**
Matching Engine

DATA & INFRA

 **PostgreSQL**

 **Redis**

 **AWS S3**

 **Kubernetes**

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Implementation Roadmap

7-MONTH PLAN & EXPECTED IMPACT

Project Plan

Phased Rollout Strategy

Total Duration: 7 Months

Phase 1: MVP Foundation

Months 1-3

Core Infrastructure Setup

- ✓ Auth & User Profiles
- ✓ Basic Matching Engine
- ✓ Database Schema

Phase 2: Integration

Months 4-5

Ecosystem Connectivity

- ✓ LMS (Moodle) Sync
- ✓ Mobile App Beta
- ✓ Matching ML v2

Phase 3: Optimization

Months 6-7

Scale & Refine

- ✓ Analytics Dashboard
- ✓ UX Audit & Polish
- ✓ Load Testing Scaling

Performance Targets



<200ms

DB Query Time



≥90%

Cache Hit Rate



70%

CPU Autoscaling

Expected Impact



Reduced Skill Dispersion

More balanced groups via variance minimization.



Increased Network Density

Better connectivity and centrality fairness.

Conclusions & Future Work

PROJECT SUMMARY & NEXT STEPS

 Final Overview



Project Achievements

Structured Transformation

The Academic Collaboration Network successfully transforms informal, chaotic group formation into a structured, measurable system. By integrating **Graph Theory** and **Algorithmic Matching**, the platform mitigates social isolation and optimizes skill distribution.



Survey Validated
Confirmed need via 25 student participants



Simulation Tested
Variance minimization proven effective



Measurable Goals



80%

ADOPTION TARGET
Conversion from WhatsApp



≥ 70

SUS SCORE
System Usability Scale



Future Work Roadmap



Pilot Deployment
Live testing with a single academic cohort.



Peer Endorsements
Skill validation system by group peers.



Video Integration
Built-in video calls for remote study.



Multi-Institution
Scaling architecture for cross-university use.